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Bayesian Inference

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INTRODUCTION

VIDEO INTRODUCTION

Optional: Watch this video on the subject: https://www.youtube.com/watch?v=HZGCoVF3YvM Another video that gives a more basic overview with more examples can be found here https://youtu.be/9TDjifpGj-k?si=T_IzGRednAJ8EPEj.

USEFUL EQUATIONS

Bayes' Theorem:

$$P(H|E) = \frac{P(E|H) * P(H)}{P(E)} \tag{1}$$

Where H is the hypothesis/model and E is the evidence/data. $P(H)$ is the prior probability (before data is taken) of H being true. $P(H|E)$ is the posterior probability or the probability of H given E. $P(E|H)$ is the probability of observing E given H. $P(E)$ is the probability only based on the data, which can be thought of as a normalization factor.

The binomial probability distribution:

$$P(h|\theta) = \binom{n}{h}\theta^h(1-\theta)^{(n-h)} \tag{2}$$

Where h is the number of heads, θ is the probability of getting a head and n is the total number of coin flips

The β function:

$$P(\alpha, \beta, \theta) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}\theta^{(\alpha-1)}(1-\theta)^{(\beta-1)} \tag{3}$$

Where $\Gamma(x)$ is the gamma function.

1. JUST THE BAYE'S-ICS [2 POINTS]

Henry is a Twitch Streamer who has built a large following by placing bets on stream. His latest hustle is having his chat bet on the outcome of his coin tosses on stream. Over the course of a stream, Henry flips a coin 7 times and gets and only 2 heads ($Beta(2, 5)$).

(a) [.5 Point]: Chat accuses Henry of using a biased coin, so he flips it a few more times and gets 3 heads and 2 tails. Update your prior to include the new data

(b) [.5 Point]: Chat still doesn't believe Henry, so he opts to do 4 final flips and expects to get exactly 2 heads and 2 tails. Calculate the binomial coefficient that would result if he got exactly 2 heads (hint, Use the beta binomial predictive distribution)

(c) [1 Point]: Using your equation from part B and the updated posterior from part a, find the exact probability of getting 2 heads in 4 attempts.

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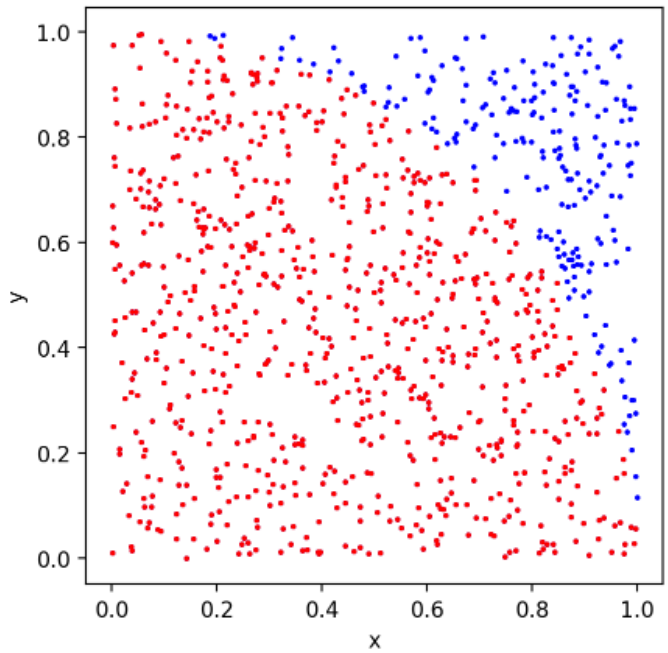

Figure 1. Showing randomized data selection for problem 3.

2. PUTTING IT IN CODE

(a) [**2 Point**]: Write a code for Bayesian inference. This code should allow inputs for the prior function, the data and the likelihood function. It should output the posterior probability distribution function. Do not use a pre-made package for the beta function.

(b) [**1 Point**]: Use your code to find a solution to a coin toss similar to problem 1, but using the provided data in data.txt. Plot your probability distribution function. Is this a balanced coin?

(c) [**1 Point**]: Test your code using different amounts of the data. Plot the distribution when 5, 50 and 500 coin flips are used. Comment on what you find.

(d) [**1 Point**]: Repeat the previous part using different prior guesses. How does your prior distribution effect the posterior distribution with different amounts of data?

3. MORE IN DEPTH

In this next section, we will calculate π using the knowledge that $\pi = \frac{A}{r^2}$.

(a) [**1 Point**]: Create a random distribution of 100 points in a square of 1×1. Find how many of these fall in a quarter of a radius 1 circle (see Figure 1). This is your data.

(b) [**1 Point**]: Using Bayesian inference, find the probability of a point falling inside of the circle. From this information and $\pi = \frac{A}{r^2}$, calculate an estimate for π.

(c) [**1 Point**]: Test how the relative error is affected by the prior and the number of data points used. Comment on what you find.